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Office of Science and Technology Policy  
Washington, DC 20024**

On the Matter of )

Request for Information on the )  
Development of an Artificial )  
Intelligence (AI) Action Plan )

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**COMMENTS OF CONNECTED NATION, INC.**

**Network Interconnection, Latency Reduction & Resilience:  
Critical Elements of a National AI Strategy**

Connected Nation, Inc. welcomes the opportunity to provide comment to the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) and the White House Office of Science and Technology Policy (OSTP) regarding the development of a national Artificial Intelligence (AI) Action Plan.

Founded in 2001, Connected Nation is a mission-driven national non-profit dedicated to ensuring that all people, regardless of location, can realize the opportunities that connectivity and technology can offer. Throughout our history, we have fostered the development of telecommunications infrastructure in unserved and underserved areas, serving as a bridge between network operators and government agencies to ensure that everyone, everywhere can get connected. The infrastructure that enables AI is the next frontier that we must tackle.

We welcome the policy pivot in favor of “permissionless innovation” represented by President Trump’s Executive Order 14179. Innovation is a strategic national security and economic development imperative. The United States must demonstrate AI leadership and dominance in all areas. The federal government’s AI policy, as outlined in EO 14179, prioritizes “removing barriers to AI innovation.” Yet a key barrier that exists today is often overlooked: network inefficiency caused by an inadequate number of carrier-neutral Internet Exchange Point (IXP) facilities across the U.S.

IXPs are physical buildings where Internet Protocol (IP) networks connect locally to exchange data traffic (i.e., peering). This happens through direct fiber cross-connects between IP networks, as well as through Virtual Local Area Network (VLAN) connections across an Internet Exchange (IX) switch, the presence of which is what distinguishes an IXP from other buildings. The lack of IXPs, relative to the U.S.’s expansive geography, increases network latency and will make the widespread deployment of latency-sensitive AI applications at the network edge

impossible to achieve. **A permissionless innovation environment can only exist in every region of the country if geographically distributed, carrier-neutral IXPs are also deployed.**

AI has already demonstrated some amazing capabilities, and it will continue to improve. While its impact is hard to quantify or forecast, it is a safe bet that important AI applications with high economic value have already emerged and are spreading. AI's role in the economy cannot be dictated by central planners; it will emerge through hands-on experimentation. Ordinary individuals, startups, and established companies will drive its discovery and practical implementation.

While the federal government should avoid premature or heavy-handed regulations that could stifle innovation with red tape, public planning is key to addressing AI's infrastructure needs. Three primary success factors deserve focused attention, and align with the administration's goals for achieving global AI leadership:

1. **Power:** AI needs a significant amount of electricity to power its computations.
2. **Networks & Interconnection:** Sufficient transport network capacity, traffic routing, and latency reduction through geographically distributed interconnection facilities, also known as IXPs. (Our focus)
3. **Security:** Keeping AI secure, resilient, on task, and immune from being hacked by bad actors.

Government has traditionally maintained a footprint in these areas due to factors like natural monopolies, environmental impacts, public land use, regulatory constraints in telephony, and network security's overlap with law enforcement. Effective policy should guide private investment through incentives while ensuring that public planning complements, not crowds out, private efforts. Policymakers need a clear vision of future AI infrastructure to guide effective decision-making.

The following comments focus primarily on *key network elements* of a vibrant AI ecosystem and particularly on the need for investment in geographically dispersed carrier-neutral IXP facilities to enable the ultra-low latency connections that generative and inference-based AI applications will require, while also fostering increased network resilience, improved data traffic routing, Layer 2 peering among network operators, and the evolution of mature marketplaces for fiber transport and IP transit competition.

For 20 years, federal connectivity policy has prioritized last-mile access and bandwidth expansion. However, reducing latency demands a different kind of infrastructure investment from both public and private sectors. While much of the focus on AI-enabling infrastructure to date has been on the creation of gigawatt-scale mega data center campuses for AI training (along with the electrical generation, transmission, and distribution needed to power it), the deficiencies in network infrastructure to enable ultra-low-latency connections locally have largely been overlooked—namely, the need for significantly more neutral points of network interconnection.

According to our own analysis of the data found on PeeringDB.com—widely considered to be the most authoritative resource on peering and network interconnection globally—57 cities in the United States have at least one IXP facility for the exchange of Internet traffic among disparate networks. Yet 14 states and three U.S. territories have no such facility at all, and three additional states have a single facility that is either failing or functionally limited. As a result,

Internet traffic from these areas must travel hundreds or even thousands of miles to the nearest peering point, adding unnecessary latency—akin to driving from Washington, DC, to Baltimore via Atlanta.

The “trombone effect” of inefficient traffic routing will certainly hamper the exchange of traffic with hyperscaler networks (i.e., AWS, Microsoft, Meta, Google, etc.) as they deploy latency-sensitive generative and inference-based AI applications at the network edge. There *must* also be neutral network interconnection facilities at the network edge—facilities that serve as a neutral venue for local Internet Service Providers (ISPs), hyperscaler networks, transport networks, mobile networks, education networks, etc. to interconnect with one another *locally*.

This model for neutral network interconnection has been the status quo in major metros for over 20 years, beginning with the evolution of neutral carrier hotels—somewhat by happenstance—in major cities like New York, Chicago, Atlanta, Dallas, and Seattle, and gradually migrating to slightly smaller metros like Washington, DC (Ashburn, VA), Kansas City, Salt Lake City, Miami, and Phoenix. These facilities are now core internet infrastructure. But with the expected rapid deployment of latency-sensitive applications, reliance on network interconnection through those cities alone is no longer an option. *The next evolution of the internet ecosystem must be planned and well-coordinated.*

China, which has a centrally planned economy and a tightly controlled technology ecosystem, is rapidly expanding state-funded AI infrastructure to ensure seamless low-latency data exchange across its AI clusters. To win the AI “race” with China, the U.S. must balance efficiency with free market principles while supporting essential infrastructure development. As with other types of shared infrastructure (e.g., interstate highways, airports, and shipping ports for the efficient routing of goods and people), there will need to be a cooperative interplay between public, private, and non-profit sectors for the efficient routing of data traffic.

President Trump highlighted the need for competitiveness in his 2019 Executive Order on Maintaining American AI Leadership, emphasizing the importance of AI infrastructure as a competitive and national security advantage. Expanding IXPs aligns with the administration’s AI leadership goals by improving real-time data processing capabilities, particularly at the network edge for AI use cases that require ultra-low latency. Without investment in high-performance IXPs, AI application such as autonomous vehicles, drone logistics, and edge computing will face unnecessary performance barriers.

Moreover, the National AI Initiative Act of 2020 (Public Law 116-283) recognizes that AI infrastructure is a strategic imperative. However much of the federal investment to date has focused on compute power rather than network interconnection and latency reduction. IXPs are a missing pillar in AI infrastructure investment that ensures ultra-low latency data transfer and strengthens AI-driven industries. Without a robust IXP network, AI innovations will be constrained by inefficient data exchange.

The Appendix below discusses promising applications such as AV-to-AV communication, drone delivery, augmented reality (AR), smart tools, robotics and AI tutors. Of course, it is very difficult to anticipate beforehand what new applied technologies will break through into mass adoption. However, the demand for high-performance connectivity has been surging for decades, and as bandwidth increases, the latency aspect of lag time reduction becomes more important. To

minimize latency will require a more distributed backbone and interconnection architecture of the internet, requiring more IXPs to be developed in more places.

Our comments below proceed as follows:

*First*, we explore the geography of AI and how the optimal AI ecosystem will distribute functions between centralized data center campuses and the near edge or “fog” layer of data centers closest to users via local IXPs.

*Second*, we discuss how public policy can promote the development of the more distributed AI infrastructure necessary to support universal or near-universal access to the cutting-edge AI applications that we anticipate will require ultra-low latency.

We conclude with a summary of our recommendations and how they align with OSTP’s priorities.

In the Appendix that follows, we put forward a few potential candidates for “killer app” use cases that may revolutionize our economy and society while relying on ultra-low-latency connectivity in ways that will make any lack of a local IXP a pain point that must be addressed.

## **The Geography of AI: Centralized Training vs. Edge AI**

The development of AI mirrors traditional human education: training first, followed by practical application.

In the training stage, large language models use machine learning to develop intelligence, drawing from the vast expanse of internet-based information, that simulates human thought. Once trained, AI applies its learned capabilities in real-world scenarios on demand. Generative and inference AI rely on this foundational training, continuously evolving and improving over time.

These two stages—training and application—*have different geographic footprints*.

Training Large Language Models (LLMs) and machine learning require immense computing power, typically centralized in rural areas where electricity is abundant and inexpensive. In these cases, network latency is not a major concern. Conversely, AI’s generative and inference tasks function like other consumer-facing applications, making them well-suited for deployment at the network edge. As models evolve, Edge AI and ‘Fog AI’ (which operates between the edge and the cloud) will likely expand, enabling broader distribution to CDNs and edge devices.

AI’s geographic distribution will continue to evolve, balancing computing power and latency. Tasks requiring extreme computing power will remain centralized, while applications demanding ultra-low latency will increasingly shift to the network edge.

Latency in AI has multiple meanings. Compute latency refers to the time an AI application takes to process a request, while network latency measures data transmission delays. Both play a crucial role in determining whether AI tasks are handled in the cloud or at the network edge.

If compute latency is substantial (e.g., several seconds), network latency below 100 milliseconds has little effect on the user experience. For example, typical network latency and bandwidth are unproblematic for text-based conversations with AI chatbots or for the basic function of using prompts to generate images. If an AI task is constrained by bandwidth limitations or is not time-sensitive, centralizing it remains a practical approach.

Edge AI adoption is most likely for time-sensitive tasks of moderate computational complexity, and for these cases, IXP proximity will become important. Ultra-low latency AI applications will likely emerge in regions already equipped for them, driven by entrepreneurial and grassroots innovation. These technologies will then expand to other low-latency regions as they evolve. However, a new latency-driven digital divide will emerge in areas that continue to lack access to a nearby IXP, with whole regions unable to support new and emerging AI-powered technologies that require ultra-low latency to function.

### **How Public Policy Can Promote Distributed IXPs for Low-Latency AI Readiness**

In the decade before 2020, a consensus grew that adequate internet service is a basic need and its universal availability needs to be a policy objective, verging on a tenet of the social contract. This eventually bore fruit in major public and private investments in last-mile broadband networks to increase bandwidth, extend it to unserved areas, and make it affordable. While these goals are laudable, public investments thus far have largely ignored reducing latency to any significant degree, which is becoming more important as bandwidth constraints ease and internet use grows more interactive and cloud-dependent.

A more intentional public policy focus and possibly to some extent, public investment, will be needed in order to tackle this emerging aspect of the digital divide and enable all Americans to benefit from cutting-edge AI applications. Achieving ultra-low latency nationwide, with public and private sector support, is estimated to cost between \$500 million to \$1 billion, far less than the BEAD Program or the announced hyperscaler investments in AI data centers.

The BEAD Program (\$42.45B), the most recent and largest in a series of federal efforts to expand broadband coverage that included the FCC's Rural Digital Opportunity Fund (RDOF) program (\$9.2B) and the ARPA Capital Projects Fund (\$10B), crowned an effort (and federal commitment) to achieve universal broadband coverage. BEAD will increase access for millions of people with *bandwidth* available to them of at least 100 Mbps downstream and 20 Mbps upstream. In many cases, this will involve the deployment of end-to-end fiber that can deliver gigabit symmetrical speeds. Unfortunately, BEAD is unambitious with respect to latency. Not only does its standard for "reliable" broadband service settle for a mere 100 milliseconds of latency, allowing greater lag time than most existing internet connections today, but it doesn't do anything to incentivize or reward less-latent connections. In this respect, its treatment of latency contrasts with bandwidth, for which the rather unambitious 100/20 minimum standard is supplemented by a preference for "priority" broadband projects with gigabit symmetric speeds. BEAD projects can be expected to reduce latency slightly since fiber will move data faster at the last mile than the copper it replaces. Some current subscribers of geostationary/high-orbit satellite internet, in particular, will enjoy huge latency reductions thanks to BEAD. But for the most part, BEAD kicks the can of latency reduction down the road.

While many internet-based applications function adequately with latencies above 100 milliseconds today, reducing latency (particularly by minimizing lengthy data backhaul and inefficient routing) will be increasingly crucial *in four key ways*.

First, the faster the bandwidth (or throughput), the more of a network's response time is driven by latency rather than bandwidth constraints. This is a matter of simple math. For a user loading a 1 MB page, a 25 Mbps connection takes 320 milliseconds due to bandwidth limits, with 100 milliseconds of latency adding 30% more delay. On a gigabit connection, bandwidth delays drop to 8 milliseconds, making latency the dominant factor, increasing load time by over 1,000%.

Second, AI will likely accelerate the shift of compute functions from local devices to powerful cloud and near-edge servers.

Third, machine-to-machine communication use cases, such as in healthcare monitoring and smart cities, are expanding. AI will further accelerate this trend by enabling adaptive devices to take on traditionally human functions. The low-latency AI use cases discussed in the Appendix illustrate these trends.

Fourth, technologies like L4S<sup>1</sup> optimize data flow management, reducing "loaded" latency compared to "unloaded" latency. This brings practical latency closer to the theoretical minimum dictated by physical data travel limits.

As computing adapts to the new opportunities created by ultra-low latency connections in many areas, a digital divide will emerge between regions that have local IXPs and regions that have higher latency because every interaction with the internet requires a data journey of hundreds of miles. The time will come when popular new applied technologies will underperform *or just won't work at all* in areas where latency is too high because data traffic is being routed through IXPs that are physically too far away.

Just as the BEAD program set a goal for universal 100/20 broadband, a national target for IXP proximity is needed to ensure full participation in the AI-powered internet. A reasonable benchmark would be deploying IXPs so that no broadband serviceable location (BSL) is more than 60 miles away: roughly the distance light travels in fiber before amplification is required. Currently, less than 20% of U.S. land is within 60 miles of an IXP<sup>2</sup>, creating significant coverage gaps in national internet infrastructure. While most IXPs are near major population centers, millions of Americans remain hundreds of miles from one, facing growing digital disadvantages as new technology leverages the median internet user's low-latency capabilities.

Of course, just because a need exists doesn't mean the government needs to address it. Often, *laissez-faire* is the best approach. IXPs are, in most cases, run as private businesses and are often profitable. Nonetheless, the economics of IXP administration are somewhat different than a traditional for-profit business because of the need to preserve neutrality.

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<sup>1</sup> See <https://potsandpansbyccg.com/2025/03/04/new-technology-to-lower-latency/>

<sup>2</sup> The basis for this calculation is that (a) 57 metros have IXPs, (b) a 60-mile radius contains just 11,000 square miles, (c) the U.S. landmass is 3.8 million square miles, and (d)  $57 * 11,000 < 20\% * 3.8 \text{ million}$ . A more sophisticated analysis would take into account that some IXP-endowed cities are coastal.

The value that IXPs create is derived from the network interconnections they enable. The full value they generate, however, is difficult to quantify in terms of service fees alone. Carrier neutrality is essential for long-term efficiency and governance, as ISPs are hesitant to build into facilities owned by or aligned with competitors—or find it prohibitively expensive to do so. Preserving a level playing field requires both rules and fair pricing, preventing IXPs from charging higher fees based on the actual derived value of service benefits. So, while successful IXP facilities often do earn a profit, they often do not attract the private capital necessary to cover the capex costs to build them because of the necessity to preserve a level playing field for network operator customers.

Therefore, the challenge for IXP economics is that a for-profit business model gets in the way of complementary investments that are needed for IXP success, especially if private investors look for high ROIs and are inclined to aggressively monetize the upside of success. Because the purpose of an IXP is interconnection, every ISP or Content Delivery Network (CDN) that connects within it raises the facility's value to other ISPs and CDNs. As these “network externalities” accumulate, an IXP gains increasing pricing power vis-a-vis the ISPs and CDNs that use it. If ISPs and CDNs anticipate that the IXP will abuse this pricing power, they may be reluctant to build into it in the first place. Therefore, it can be important to IXP success that they operate at least partly with the public-interest objectives of 1) reducing latency for network customers and end-users, 2) improving regional network resiliency, and 3) lowering wholesale prices—all with a long-term horizon for measuring success in these areas.

Public investment in closing IXP coverage gaps follows the same core rationale (and challenges) as BEAD's last-mile subsidies. It addresses local market failures by filling financial gaps where revenue capture alone is insufficient. It also promotes fairness, ensuring communities aren't left behind due to geography. Additionally, it helps build the market scale needed for cutting-edge innovation, freeing technologists to advance solutions without backward-compatibility constraints. However, like BEAD, directing public funds into private infrastructure that serves both public and private interests presents complex legal, institutional, and decision-making challenges that require careful navigation.

That is why it's important to set goals early. It may be several years before AI-powered technologies that require ultra-low latency achieve mass adoption to the point where the infeasibility of using them in regions lacking IXP proximity becomes an acute pain point of digital disadvantage. By that time, however, the opportunity to proactively solve the problem will have been missed, and investment in fixes is likely to be either hasty and inefficient or slow enough to do considerable cumulative economic damage while the area waits for a solution. ***The time is now to get ahead of the problem.***

Funding is not a major barrier, as the financing gap for IXPs is minimal compared to federal agency and hyperscaler budgets. In some cases, well-designed policies to encourage hyperscaler IXP co-builds can eliminate the need for public investment. Where public funding is needed, a few million dollars per IXP, or less than a billion dollars nationally, should suffice to meet ambitious latency reduction goals.

The BEAD Program is expected to have hundreds of millions (possibly billions) in “non-deployment” funds remaining after the program's last-mile broadband objectives have been met. States anticipate using these funds locally rather than returning them to the federal government.

Efforts to modify BEAD, including the SPEED for BEAD Act, seek to limit how these funds can be spent. The NTIA, within existing statutes, should redirect funds toward broadband infrastructure like IXPs instead of non-infrastructure initiatives that will not have a long-term impact. Small but strategic adjustments to BEAD could drive meaningful investment in geographically distributed carrier-neutral IXPs.

Universal Service Fund reform, which has long been perceived as an acute need by the telecommunications industry and may be necessitated by an imminent Supreme Court decision, could be designed in ways that would facilitate and support IXP investment. Optimizing the funding solution, either with new legislative appropriations or by leveraging any of the wide variety of existing programs, is not trivial but is downstream of framing national policy goals.

As a final consideration, in disaster scenarios, such as an Electromagnetic Pulse (EMP) event from solar flares or a nuclear attack, geographically dispersed IXPs would significantly improve internet resiliency—especially if newly constructed IXPs are hardened against EMP events. The transition from copper (highly vulnerable to EMPs) to fiber (which is not) has reduced some risks, but connectivity remains dependent on the power grid and the electronics used to light fiber optic cables at each end. Communities that are dependent upon IXPs, even those far beyond the direct impact zone, will see their connectivity disrupted for long periods of time, even if local power remains intact. Investing in more geographically distributed IXPs would not only lower latency under normal conditions but also ensure the internet remains functional in the wake of disasters. As AI further integrates into daily life, strengthening network resiliency becomes even more crucial.

### **Conclusion: A National AI Action Plan Must Not Overlook the Crucial Need to Reduce Latency and Enhance Network Resilience**

In summary, Connected Nation supports the creation of a national AI action plan that addresses the need for infrastructure development beyond just power and security. Latency reduction and enhanced network resilience must lie at the core of any strategy. We believe these challenges can only be overcome through the creation of physical network interconnection facilities, commonly known as Internet Exchange Points, that are geographically distributed, carrier-neutral, and hardened to withstand natural and man-made disasters.

To achieve the country's national security and economic competitiveness objectives, any national AI strategy should adhere to three principles that OSTP has emphasized in its own planning: 1) all Americans deserve the opportunity to benefit from AI advancements, 2) national security requires AI resilience, and 3) America must assert global leadership in AI and remain dominant in the race against China.

Just as AI computing resources must be distributed nationally, AI network interconnection must also be decentralized. Today, 14 states and 3 U.S. territories lack neutral IXPs. This forces AI related data traffic to be routed inefficiently. Any federal AI strategy should prioritize neutral IXP deployment as a core infrastructure investment, ensuring that all Americans benefit from the innovation and economic opportunity that AI will bring.

It is no secret that AI-enabled defense and intelligence applications require secure real-time data exchange between systems. The problem: current IXP gaps create single points of



failure, making AI systems vulnerable to network disruptions, EMP, cyberattacks, and foreign interception. The National Security Commission on AI has warned that U.S. digital infrastructure is a weak point in AI deployment. Without a distributed IXP ecosystem, real-time AI applications such as cyber defense, autonomous systems, and AI-assisted intelligence analysis face avoidable performance risks. The federal government must recognize neutral IXPs as critical infrastructure to support AI resilience.

The U.S. and China are in an AI infrastructure race, with China aggressively investing in data centers, cloud computing, and high-speed network interconnection. Unlike China's state-coordinated AI infrastructure, the U.S. has no national policy for AI-related network optimization, leaving key regions of the country with poor latency performance and many single points of failure. A national target for carrier-neutral IXP proximity is needed, and existing and future IXPs should be classified as critical infrastructure.

We commend the administration for its efforts to develop a comprehensive national AI action plan, and we welcome any opportunity to engage further on this important topic.

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## **Appendix**

### **Ultra-Low-Latency AI Use Cases**

Below are key potential use cases for AI-driven applications requiring ultra-low latency connectivity.

#### **1. Autonomous Vehicle-to-Autonomous Vehicle (AV-to-AV) Communications for Traffic Safety**

Autonomous vehicles (AVs) will play a transformative role in the economy, revolutionizing logistics and transportation. The use-cases are numerous: more efficient grocery delivery, streamlined peer-to-peer sales (e.g., negotiated via Facebook Marketplace and delivered by AV), more mobility for seniors, teenagers, and the disabled, less car ownership (with benefits for many people's personal finances), fewer traffic collisions, a repurposing of urban parking lots, and near-limitless other benefits. Advances in AI are finally making widespread driverless adoption a reality. In cities like San Francisco, Austin, Phoenix, and Miami, commercially operated robotic taxis are already on the streets.

Driving demands *quick reflexes*, whether human or machine—requiring rapid, situationally informed decisions. In many cases, split-second accuracy can mean the difference between life and death.

AVs don't always require ultra-low latency communications for driving decisions because they require significant onboard computing power. AVs cannot be cloud-dependent to the point of losing functionality during network disruptions. However, this computing power has significant financial and design implications—adding thousands of dollars to the car's price, requiring a couple cubic feet of space, and weighing up to 50 pounds. This may be acceptable in something as large as a car but completely impractical for many other machines.

In hazardous situations, AV-to-AV communication will be a crucial ultra-low latency use case. While currently less relevant due to AVs' small market share, its importance to traffic safety will grow as adoption increases.

Of course, vehicle-to-vehicle (or driver-to-driver) communication, e.g., via brake lights and turn signals, is important to traffic safety even now. But the set of signals drivers can send to one another is very limited. AV-to-AV communication can be much richer but will probably require the cloud, and its need for real-time, situational decision-making is intolerant of delay.

Consider a hazardous scenario in which a child runs into a road as two AVs approach from opposite directions. Other children are playing near the roadside from which the child came. There is insufficient time to brake to a stop. The AV in the nearer lane can swerve to avoid the child, but if the other AV stays its course, that will result in a head-on collision, likely fatal to the passengers in both AVs. However, if the other AV *also* swerves, steering into the far shoulder of the road, a collision can be prevented, and all the human lives involved can be saved. Human drivers typically could not execute such a maneuver in real time. AVs have that potential, but extremely rapid communication between the two AVs would be necessary to coordinate the maneuver.

AV-to-AV communication has many other possibilities. For example, it is energy-efficient for cars to convoy, one following close behind another so that each one can benefit from the

slipstream of the last. This is currently prevented by traffic rules for safety reasons, since it is highly likely that, if one driver had to brake suddenly, the next driver's reflexes would not be fast enough to avoid rear-ending the car ahead. With good AV-to-AV communication, convoys could coordinate braking behavior and reduce or eliminate collision risks. Such convoys could also make more efficient use of scarce road space.

Again, the evolution of traffic patterns and driving decisions to leverage ultra-fast AV-to-AV communications is most likely to occur largely in places that already have local IXPs and low latency. When it does, places lacking local IXPs may be unable to adopt emerging best practices. This form of digital disadvantage could get dangerous if carmakers adapt their AVs to rely on ultra-low latency AV-to-AV communications only to discover that their AVs can't operate properly in high-latency areas. It might be necessary to limit the use of cutting-edge AVs to low-latency areas.

## **2. Drone Delivery, as Enabled by Cloud-Based Real-Time Decision-Making**

A second candidate use case for ultra-low latency is drone delivery. Drones resemble AVs in many ways, with the key difference being onboard computing power is far more of an operational problem for a drone versus a car. Drones need to outsource more computing and decision-making to servers at the network edge.

Drone delivery is less intuitive than AVs, which mimic human functions. Drones have been operationalized, and their economic promise is enormous. Delivering small packages by drone rather than by car is inherently cost-effective. Drones are far smaller and faster. While cargo drones' flight velocity tends to be like that of cars, they are often faster because they can follow straighter courses and face no stop signs, stoplights, etc. Drones can also go to places that cars can't, such as islands, boats on the water, hikers in the forest, or the balcony of an apartment on the tenth floor.

Widespread drone delivery could transform lifestyles. While not a full substitute for ground-based transportation, drones will enable deliveries to remote locations, especially when combined with off-grid technologies like LEO satellite broadband, solar panels, and batteries—potentially allowing teleworkers to live in truly remote areas.

If it becomes generally available, drone delivery will enable new lifestyles. Currently, modern life is largely infeasible outside the immediate vicinity of a road traversable by car. While drone delivery will not be a comprehensive substitute for car access, the option of drone delivery from retailers and restaurants to roadless points on the map, combined with other emerging off-the-grid technologies like LEO satellite broadband service, solar panels and batteries, opens up fascinating possibilities for teleworking professionals to subsist indefinitely in wild beauty spots such as islands. At the other end of the urbanism spectrum, city dwellers might be able to order goods by drone delivery while out and about or to penthouse apartment balconies while staying in. Benefits to suburban single-family home dwellers start with fewer trips to the store.

Like AVs, delivery drones, which need to operate Beyond Visual Line of Sight (BVLOS), need to make real-time decisions, such as rerouting in the event of unexpected obstacles, performing detect-and-avoid maneuvers, adapting to wind conditions, and finding good spots to

land or drop cargo when they reach their destinations. As with AVs, this decision-making will often need to be very rapid. But there are several big differences between AVs and drones:

1. AVs need to integrate into a well-developed traffic system designed around human drivers, which may later be optimized for AVs. For drones, the traffic system needs to be created.
2. AV operations always pose a danger to human life which must be minimized or eliminated by good driving decisions. Drones don't ordinarily jeopardize human life but can damage buildings and infrastructure.
3. AVs must operate in the road grid, which is two-dimensional and circumscribed, whereas drones can use three-dimensional space and so have far more room to operate.
4. AVs can afford far more onboard computing power, whereas drones will need to rely on the cloud or, preferably, on servers at the network edge, via ultra-reliable low-latency communications (uRLLC).

Drone detect-and-avoid systems will initially need to deal more with natural obstacles such as trees, telephone lines, tall buildings, birds and bats than with other drones. For that, sensors and the interpretation and use of their data are key. As drone fleets grow, the risk of drone-on-drone collisions will increase, and drone-to-drone communications, probably via servers at the network edge, will increase in importance.

Geographically, AVs and drone delivery may differ because road density relative to land and population has, in some ways, opposite effects on demand. AVs need roads and will thrive where road density is high and roads are good. By contrast, drones *don't* need roads, and drone delivery may sometimes find the most eager adoption in rural and remote areas where roads are few.

The risk that early drone delivery systems will pose to ground and vertical structures will be far lower in rural and remote areas, where an errant drone will mostly hit trees, than in urban centers, where they may crash into buildings or power lines. While urban drone delivery operations could create extremely high value, it may take a long time before their safety track records are strong enough to induce the densest of metropolitan centers to open their airspace to high-density commercial drone delivery operations. Rural and remote areas are ideal venues for drone delivery to start initial operation and accumulate a track record that can justify operations in urban airspaces later on; however, this only works if the rural and remote areas can offer the infrastructure, such as nearby IXPs and ultra-low latency, that make drone delivery operations possible.

### **3. AI Invades Physical Space: Augmented Reality, Smart Tools, and Robotics**

Until now, AI has primarily existed in the digital realm (text, images, and code) but will soon expand into physical space, driven by the convergence of:

1. *Augmented reality (AR)*. Human functionality can be enhanced by AR tools that add relevant data to the human field of vision by means of wearable devices.
2. *Internet of things (IOT)*. Many objects can be enhanced by connectivity to the cloud, emitting and/or receiving information.
3. *Robotics*. Adaptable machines can imitate human motor functions and substitute human labor for a variety of tasks.

Despite their impact in industry, AR, IoT, and robotics have struggled with mass consumer adoption, largely due to unintuitive human-machine interfaces. This applies outside the realm of information technology, too. Tool makers have long faced the challenge of making their products user-friendly, and a key barrier to adding features that customers might value is that customers may never figure out how to use them. User manuals are notoriously underused.

This is where AI promises to be transformative. AIs excel at making information intuitive and understandable. A tool whose features are explained by an AI chatbot rather than an old-fashioned user manual has immediate appeal. But combining this with AR yields even more exciting possibilities. An AR headset could, for example, label the features of a newly purchased tool with markups to the human field of vision and respond to voice questions from the user about what features do. This is already being done in industrial and healthcare settings, and AI could facilitate mass adoption by making the technology more user-friendly.

Two new frontiers could be opened by AI-powered AR instruction in tool use.

First, tools themselves can become more complex with more features. AI might explain the tool to users, or it might give commands to tools itself based on its understanding of what the human users say they want. That would require the tool to be online, thus integrating AR with IOT.

Second, AR and IOT functionalities powered by AI can facilitate tool *rental* as a way for households to get access to useful tools without the costs and complexities of purchase, storage, maintenance and repair. Tool rental is an active yet somewhat underutilized market, and households both routinely own tools that they very rarely use and hire professionals to do things that could be done DIY with the right tools and a few simple skills. AI can help people learn or control tools as a substitute both for tool ownership and for hiring professionals, while also enabling tool rental providers to track what happens with their rental tools and charging less to careful and competent tool renters.

Such trends would also pave the way for increased utilization of robots. They are advanced tools, with the Roomba being a prime example of home automation. While millions have been sold, household robotics still lag behind industrial automation. As AI makes robots more intelligent, able to adapt to their environments, and easier for humans to interface with them, that may change.

All of this is deliberately abstract and noncommittal about where mass consumer applications of AI in physical space, outside of transportation, will achieve breakthroughs. However, much of the functionality here will depend on ultra-low latency. For example:

- In order for an AR headset to mark up a human field of vision for greater intelligibility, the display will need to adapt to human movements extremely quickly. A few milliseconds of delay, would substantially lessen the usefulness of the AR. (Such as requiring a user to hold still for the labels to settle into the right places).
- In tool use, errors can do damage fast. An AI might be able to stop a tool in time to prevent damage, but only if its “reflexes” are good enough.
- A robot navigating its environment needs to process sensor data and model its environment in real time.

As the footprint of AI in physical space grows through AR, IOT, and robotics, low-latency areas will have an advantage. IXP investment can both save areas from being left behind and expand the addressable market for these exciting AI use cases.

#### **4. Real-Time Interaction with AI Tutors**

Anyone who wishes to can already use Bing Copilot, Google Gemini, or Grok as a kind of AI tutor. Ask questions, get answers, and learn. But to do that, one needs to be able to read, type, and be somewhat articulate in formulating questions.

Young children and early readers need a different kind of assistance. Reliance on spoken conversation rather than text enables new possibilities for AI to teach people who are not yet literate. The range of knowledge that needs to be readily accessible for an AI tutor teaching young children (e.g., how to form letters and numbers, basic grammar and math) is far more limited than what is needed for an AI that contributes at the cutting edge of business, technology, and culture. But to maximize success, the interactive experience should be as human-like as possible.

High latency can be an impediment to that. Interactions don't "feel" natural if there is a long delay. That can impede communication because normal language processing relies on a certain reaction time. While 5-10 milliseconds of delay aren't noticeable *per se*, they add up, and some educational uses of AI may turn out to warrant decentralization to the network edge for the sake of optimal response times.